

India's Agricultural Crop Production Analysis using Tableau

Domain	Data Analytics	Internship	SmartBridge – SkillWallet
Dataset	Indian Agricultural Crop Production Dataset (1997–2021)	Dataset Size	~345,408 rows, 10 columns (~28 MB)
Visualization Tool	Tableau Public	Maximum Marks	4 Marks

1. FUNCTIONAL REQUIREMENTS

Following are the functional requirements established for the proposed visual analytics dashboard and story structure.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion & Preprocessing	- Import the agricultural production dataset into Tableau Public. - Perform data cleaning and handling of missing fields. - Create an optimized calculated field for Yield using the custom metric formula: Yield = Production / Area.
FR-2	Descriptive Summary & KPIs	- Generate summary KPI cards tracking total core production and area metrics. - Partition crop production distribution based on relevant volumetric unit types.
FR-3	Geospatial Analytics	- Display an interactive state-wise agricultural production map layer. - Highlight the top leading regional agricultural producers in a clear ranking matrix.
FR-4	Comparative Matrix	- Compare total cultivated geographic area partitioned by individual crops. - Display side-by-side state-wise total cultivated area comparisons. - Show the distinct season-wise agricultural production distribution share.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-5	Temporal Trend Charts	• Build a continuous year-wise overall production trend line (1997–2021). • Build a separate chronological timeline showcasing long-term crop yield trends.
FR-6	Dashboard Design	• Connect structural sheets using standard dynamic filtering parameters. • Assemble visual worksheets into a single unified Tableau Dashboard.
FR-7	Storyboarding & Delivery	• Construct a sequential Tableau Story layout to guide analytical users. • Publish the completed Tableau Dashboard and Story to Tableau Public for easy public access and visualization sharing.

2. NON-FUNCTIONAL REQUIREMENTS

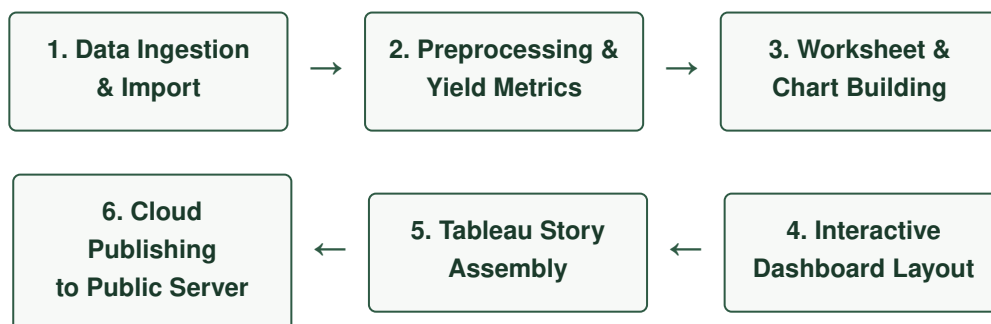
The following non-functional requirements ensure that the Tableau dashboard provides efficient performance, usability, reliability, and accurate agricultural data visualization.

NFR No.	Non-Functional Requirement	Description
NFR-1	Performance	Cross-filtering queries and charts must update within 2–3 seconds when processing the ~345,408 rows.
NFR-2	Reliability	Aggregations, tooltips, and map polygons must render consistently without breakdown during heavy usage.
NFR-3	Scalability	The workbook framework must seamlessly handle supplemental records or future year updates beyond 2021.
NFR-4	Accuracy	Calculated measures and visual distributions must exactly mirror standard algebraic statistical standards.
NFR-5	Usability	The layout must be clean and highly legible for farmers, researchers, and government analysts alike.
NFR-6	Maintainability	All dimensions, parameters, and expressions must use clean naming conventions for easy future updates.
NFR-7	Availability	

NFR No.	Non-Functional Requirement	Description
		The published Tableau Dashboard and Story will be accessible through Tableau Public whenever an internet connection is available.
NFR-8	Security	Basic data view parameters apply; underlying administrative configuration fields must remain write-protected.
NFR-9	Compatibility	Visual reports must render correctly across standard modern web browsers including Chrome and Edge.

3. REQUIREMENT FLOW DIAGRAM

The operational sequence below highlights the data roadmap from raw files to public presentation:



4. SYSTEM INFRASTRUCTURE REQUIREMENTS

The following hardware and software specifications guarantee smooth performance during regional data compilation.

Hardware Parameter	Requirement Specification
Processor	Intel Core i3 processor or higher equivalent architecture
RAM	Minimum 4 GB volume (8 GB capacity is highly recommended for faster data processing)
Storage	At least 500 MB of clear dedicated local disk storage capacity space
Display Resolution	Standard 1366 x 768 pixels layout workspace resolution or superior quality
Internet Connection	Internet connectivity is required to publish dashboards and stories to Tableau Public and to access the published visualizations.

Software Application	Allocated Operational Purpose
Operating System	Microsoft Windows 10 / Windows 11 platform architecture environment
Primary BI Tool	Tableau Public Desktop software workspace instance
Spreadsheet Tool	Microsoft Excel suite for basic tabular data previewing and tracking
Web Browser	Google Chrome or Microsoft Edge for live public deployment checks

5. INTENDED AUDIENCE & USER REQUIREMENTS

This descriptive data system caters to several key user profiles across academic, economic, and operational domains:

- **Farmers & Local Agricultural Officers:** To evaluate local regional cultivation histories and choose robust crops.
- **Government Officials & Policy Makers:** To monitor macro-level food production metrics and coordinate regional supply allocation.
- **Agricultural Researchers & Data Analysts:** To look for patterns over the multiple years of agricultural production data (1997–2021) to study climate or production shifts.
- **Students & Academic Educators:** To serve as a baseline model for demonstrating real-world spatial-temporal analysis pipelines.

6. EXPECTED SYSTEM CAPABILITIES

Once finalized, the analytical system will deliver the following strategic capabilities:

- **Interactive Dynamic Dashboards:** Seamlessly updates all associated visual sheets upon user filter interaction.
- **Story Presentation Layouts:** Guides users through step-by-step sequential crop performance insights.
- **State-wise Agricultural Production Maps:** Offers instant geographic awareness of crop distribution metrics across individual states.
- **Enhanced Decision Metrics:** Empowers agricultural analysts with clean, robust empirical validation to back up strategic planning.

7. CONCLUSION

The identified functional and non-functional requirements ensure that the Tableau solution delivers accurate, interactive, and user-friendly agricultural data analysis. By integrating KPI cards, state-wise maps, trend analysis, filters, dashboards, and Tableau Stories, the system enables users to understand agricultural production patterns from 1997–2021 and supports better data-driven decision-making. These requirements provide a strong foundation for an efficient and reliable agricultural analytics solution.